# Kinematic Slope Stability Toolkit

*Kinematic Feasibility Analysis of Rock Slopes Using the Markland Test and Goodman-Bray Criteria*

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### Project Links

**GitHub Repository:** *https://github.com/aayushkumarlal50-dev/kinematic-slope-stability-toolkit*

**Live Demo (Streamlit Community Cloud):** *https://kinematic-slope-stability-toolkit-rtegf3xfanwopzckqjfabb.streamlit.app/*

## Abstract

The Kinematic Slope Stability Toolkit is an interactive Python application that screens rock slopes for the geometric possibility of plane, wedge, and toppling failure. Given the orientation of the slope face, one or more discontinuity sets, and the friction angle of the rock mass, the tool applies the Markland test for plane and wedge modes and the Goodman-Bray criteria for toppling mode, and visualizes the result on a Schmidt equal-area stereonet, plotting joint poles, the friction cone, and the relevant failure zones.

The tool is deliberately scoped to kinematic feasibility. It does not calculate a factor of safety, does not model water pressure, and does not predict when, or whether, a slope will actually fail. For each discontinuity set it answers one well-defined question: is this failure mode geometrically possible, given the slope and joint geometry? This makes it a natural first step in a slope design workflow, intended to precede limit-equilibrium or numerical analysis wherever the screening flags a credible failure mode.

## Technical Background

The toolkit performs kinematic screening of rock slopes using two established geometric checks. For plane and wedge failure, it applies the Markland test on a Schmidt stereonet: a potential sliding surface must daylight on the slope face and also lie outside the friction cone. For wedge failure, the line of intersection of two discontinuity planes is checked in the same way. For flexural toppling, the toolkit uses the Goodman-Bray criteria, which identify steep discontinuities oriented roughly sub-parallel to the slope dip direction that can form columns capable of toppling. The sliding and toppling checks are kept separate because a discontinuity set may be critical for one failure mode and not the other.

## Key Features

- Plane failure analysis using the Markland test (daylighting + friction cone)
- Wedge failure analysis using the line of intersection of two discontinuity sets
- Flexural toppling analysis using the Goodman-Bray criteria
- Schmidt equal-area stereonet visualization of joint poles, the friction cone, and failure zones
- Support for multiple discontinuity sets analyzed together in a single run
- Interactive Streamlit interface - no coding required to run an analysis

## Methodology

### Inputs

- Slope face orientation: dip and dip direction
- One or more discontinuity sets: dip and dip direction for each set
- Friction angle (φ) of the discontinuities, in degrees

### Processing Pipeline

1. Convert slope and discontinuity orientations into stereonet poles and unit vectors.
2. Check each discontinuity set against the Markland test and Goodman-Bray criteria; for wedge mode, compute the line of intersection of each relevant pair of planes first.
3. Plot the joint poles, friction cone, and failure envelopes on a Schmidt equal-area stereonet using mplstereonet.

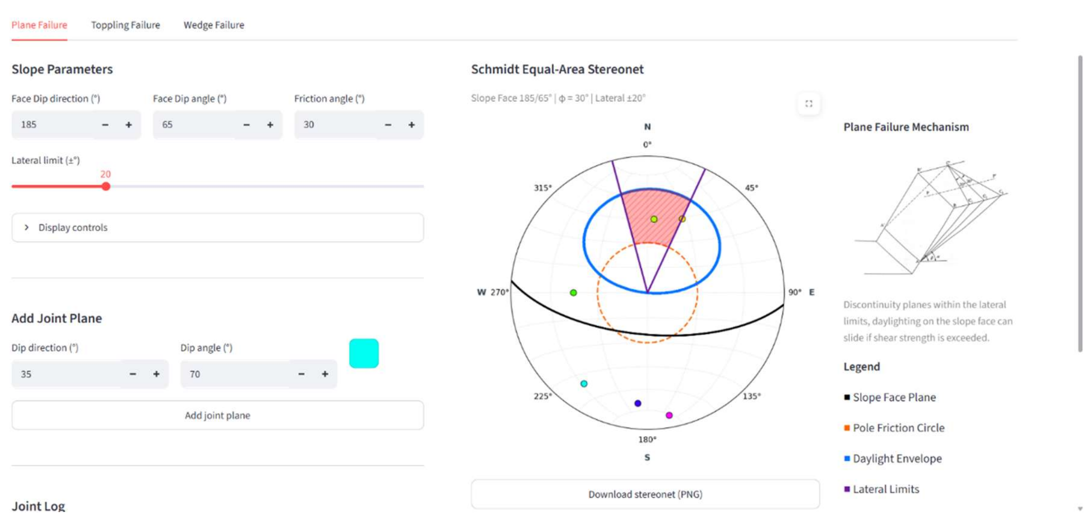
### Outputs

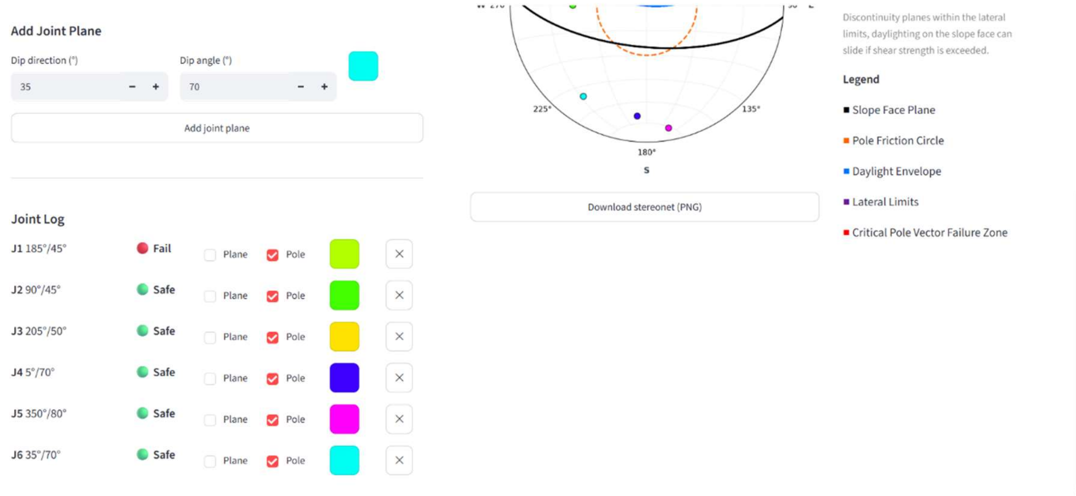
- A kinematically-critical / not-critical result for each discontinuity set, for each applicable failure mode
- An annotated Schmidt stereonet showing joint poles, the friction cone, and shaded failure zones

## Screenshots

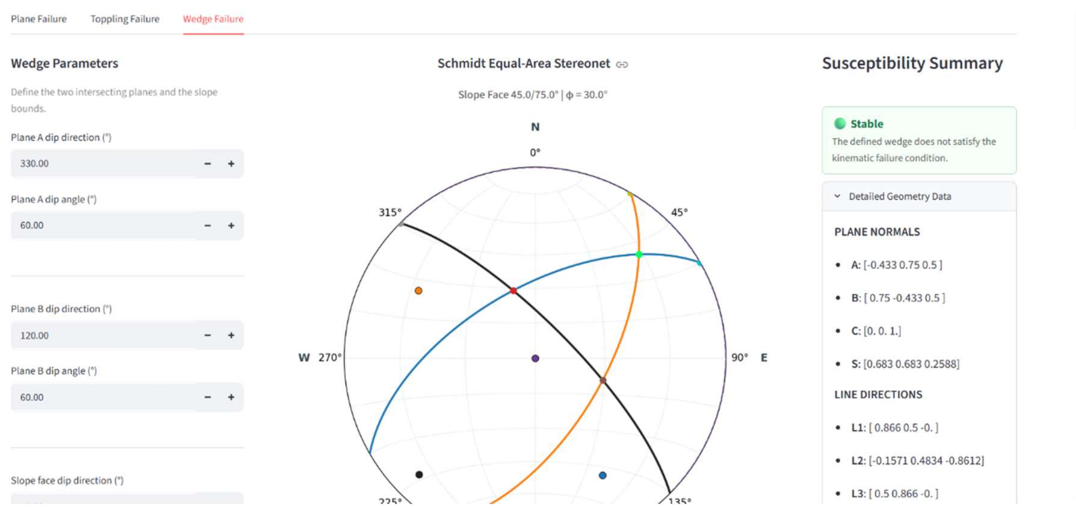
Screenshots below illustrate the app interface and a representative result for each failure mode.

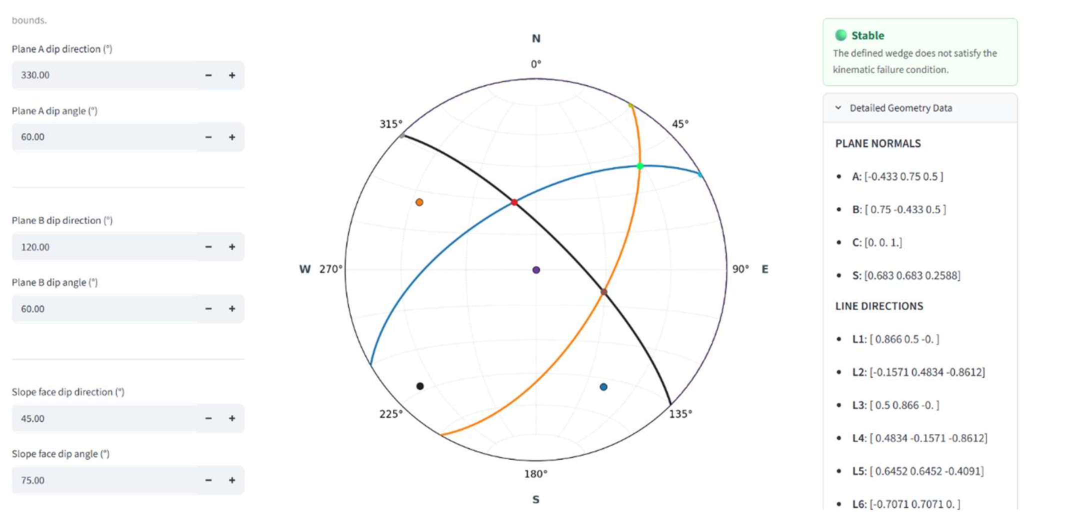
### Plane Failure



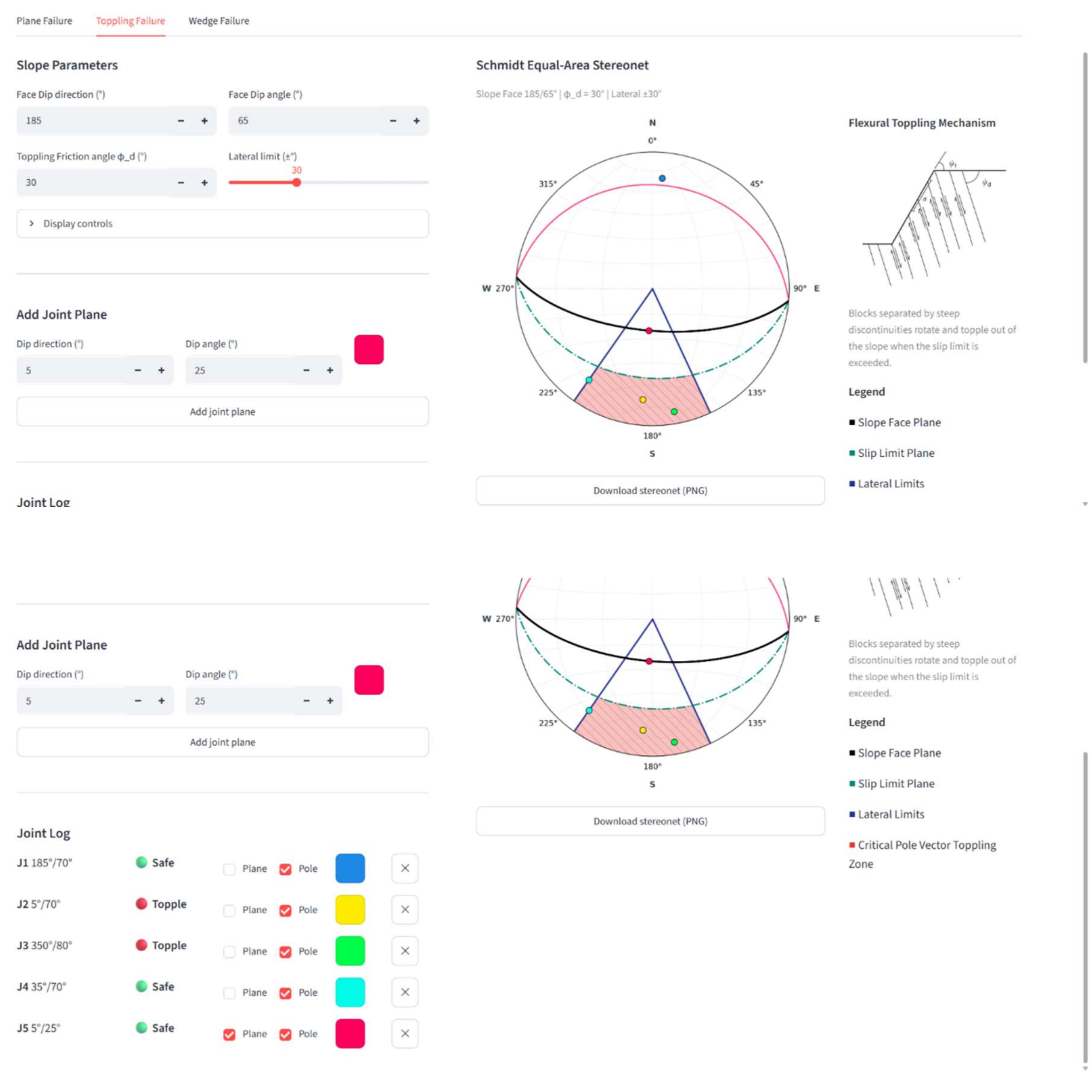



## Wedge Failure



## Toppling Failure




# System Architecture

## Technology Stack

- Python 3.8+
- NumPy: vector and angle computations
- Matplotlib: plotting backend
- Mplstereonet: stereographic (Schmidt net) projections
- Streamlit: interactive web interface

## Codebase Structure

| File | Role |
|---|---|
| `app.py` | Streamlit interface and overall app flow |
| `src/geometry.py` | Vector and angle utilities |
| `src/mechanics.py` | Markland and Goodman-Bray failure-mode checks |
| `src/tetrahedron_logic.py` | Wedge geometry engine (line-of-intersection calculations) |
| `src/visualization.py` | Schmidt stereonet plotting |

## Assumptions and Scope

This toolkit performs kinematic screening of rock slopes under simplified geometric and mechanical assumptions. It treats the rock mass as rigid blocks with planar discontinuities and uses a single uniform friction angle under dry conditions. The results indicate whether plane, wedge, or toppling failure is geometrically possible; they do not compute factor of safety, failure timing, or progressive failure, and should be used only as a first-pass screening step before detailed analysis.

> **Note:** A positive result means a failure mode is geometrically possible under the stated assumptions - not that failure is certain or imminent. Treat the output as a screening result, to be followed by limit-equilibrium or numerical analysis with site-specific data before any design decision is made.

## Author and Contact

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